ASSIGNMENT 3

Q1)

```
mysql> select * from employee;
+--------+----------+------------+----------+---------+
| emp_id | emp_name | emp_salary | emp_city | dept_id |
+--------+----------+------------+----------+---------+
|      1 | Joey     |      47300 | NJ       |    NULL |
|      2 | Peter    |      60000 | NYC      |    NULL |
|      3 | Danial   |      55000 | LA       |    NULL |
|      4 | Quagmire |     120000 | NYC      |    NULL |
|      5 | Stewie   |      62000 | NYC      |    NULL |
+--------+----------+------------+----------+---------+
5 rows in set (0.04 sec)

mysql> select sum(emp_salary) as total_salary from employee;
+--------------+
| total_salary |
+--------------+
|       344300 |
+--------------+
1 row in set (0.00 sec)
```

Q2

```
mysql> select emp_city, avg(emp_salary) from employee GROUP BY emp_city;
+----------+-----------------+
| emp_city | avg(emp_salary) |
+----------+-----------------+
| NJ       |      47300.0000 |
| NYC      |      80666.6667 |
| LA       |      55000.0000 |
+----------+-----------------+
3 rows in set (0.01 sec)
```

Q3

```
mysql> select * from employee;
+--------+----------+------------+----------+---------+-----------+
| emp_id | emp_name | emp_salary | emp_city | dept_id | dept_name |
+--------+----------+------------+----------+---------+-----------+
|      1 | Joey     |      47300 | NJ       |    NULL | SWAT      |
|      2 | Peter    |      60000 | NYC      |    NULL | NYPD      |
|      3 | Danial   |      55000 | LA       |    NULL | SWAT      |
|      4 | Quagmire |     120000 | NYC      |    NULL | NYPD      |
|      5 | Stewie   |      62000 | NYC      |    NULL | NYPD      |
+--------+----------+------------+----------+---------+-----------+
5 rows in set (0.00 sec)

mysql> select dept_name, count(emp_name) from employee group by dept_name;
+-----------+-----------------+
| dept_name | count(emp_name) |
+-----------+-----------------+
| SWAT      |               2 |
| NYPD      |               3 |
+-----------+-----------------+
2 rows in set (0.00 sec)
```

Q4

```
mysql> SELECT dept_name, COUNT(emp_name)
    -> FROM employee
    -> GROUP BY dept_name
    -> HAVING COUNT(emp_name) > 2;
+-----------+-----------------+
| dept_name | COUNT(emp_name) |
+-----------+-----------------+
| NYPD      |               3 |
+-----------+-----------------+
1 row in set (0.00 sec)
```

Q5

```
mysql> select dept_name, count(*) as total_employees
    -> from employee
    -> group by dept_name
    -> having count(*) > 1;
+-----------+-----------------+
| dept_name | total_employees |
+-----------+-----------------+
| SWAT      |               2 |
| NYPD      |               3 |
+-----------+-----------------+
2 rows in set (0.00 sec)
```

Q6

```
mysql> SELECT SUM(emp_id) as total_emp ,MAX(emp_salary) AS highest_salary
    ->
    ->
    ->
    -> from employee;
+-----------+----------------+
| total_emp | highest_salary |
+-----------+----------------+
|        15 |         120000 |
+-----------+----------------+
1 row in set (0.00 sec)
```

Q7

```
mysql> select *
    -> from employee
    -> where emp_name LIKE '%e';
+--------+----------+------------+----------+---------+-----------+
| emp_id | emp_name | emp_salary | emp_city | dept_id | dept_name |
+--------+----------+------------+----------+---------+-----------+
|      4 | Quagmire |     120000 | NYC      |    NULL | NYPD      |
|      5 | Stewie   |      62000 | NYC      |    NULL | NYPD      |
+--------+----------+------------+----------+---------+-----------+
2 rows in set (0.01 sec)

mysql>
```

Q8

```
mysql> SELECT * from employee
    -> where emp_name LIKE 'J%';
+--------+----------+------------+----------+---------+-----------+
| emp_id | emp_name | emp_salary | emp_city | dept_id | dept_name |
+--------+----------+------------+----------+---------+-----------+
|      1 | Joey     |      47300 | NJ       |    NULL | SWAT      |
+--------+----------+------------+----------+---------+-----------+
1 row in set (0.00 sec)
```

Q9

```
mysql> select *
    -> from employee
    -> where emp_name LIKE '%ia%';
+--------+----------+------------+----------+---------+-----------+
| emp_id | emp_name | emp_salary | emp_city | dept_id | dept_name |
+--------+----------+------------+----------+---------+-----------+
|      3 | Danial   |      55000 | LA       |    NULL | SWAT      |
+--------+----------+------------+----------+---------+-----------+
1 row in set (0.00 sec)
```

Q10

```
mysql> SELECT *
    -> FROM employee
    -> ORDER BY emp_salary DESC;
+--------+----------+------------+----------+---------+-----------+
| emp_id | emp_name | emp_salary | emp_city | dept_id | dept_name |
+--------+----------+------------+----------+---------+-----------+
|      4 | Quagmire |     120000 | NYC      |    NULL | NYPD      |
|      5 | Stewie   |      62000 | NYC      |    NULL | NYPD      |
|      2 | Peter    |      60000 | NYC      |    NULL | NYPD      |
|      3 | Danial   |      55000 | LA       |    NULL | SWAT      |
|      1 | Joey     |      47300 | NJ       |    NULL | SWAT      |
+--------+----------+------------+----------+---------+-----------+
5 rows in set (0.00 sec)
```

Q11

```
mysql> select dept_id from employee
    -> order by dept_id asc;
+---------+
| dept_id |
+---------+
|       1 |
|       2 |
|       2 |
|       3 |
|       3 |
+---------+
5 rows in set (0.00 sec)

mysql> select emp_salary from employee
    -> order by emp_salary desc;
+------------+
| emp_salary |
+------------+
|     120000 |
|      62000 |
|      60000 |
|      55000 |
|      47300 |
+------------+
5 rows in set (0.00 sec)
```

Q12

```
mysql> select emp_salary from employee
    -> where emp_salary between 30000 AND 60000;
+------------+
| emp_salary |
+------------+
|      47300 |
|      60000 |
|      55000 |
+------------+
3 rows in set (0.00 sec)
```

Q13

```
mysql> select * from employee
    -> where dept_id in (1,2);
+--------+----------+------------+----------+---------+-----------+
| emp_id | emp_name | emp_salary | emp_city | dept_id | dept_name |
+--------+----------+------------+----------+---------+-----------+
|      5 | Stewie   |      62000 | NYC      |       1 | NYPD      |
|      2 | Peter    |      60000 | NYC      |       2 | NYPD      |
|      4 | Quagmire |     120000 | NYC      |       2 | NYPD      |
+--------+----------+------------+----------+---------+-----------+
3 rows in set (0.00 sec)
```

Q14

```
mysql> select emp_name, emp_salary
    -> from employee
    -> order by emp_salary ASC
    -> limit 1;
+----------+------------+
| emp_name | emp_salary |
+----------+------------+
| Joey     |      47300 |
+----------+------------+
1 row in set (0.04 sec)
```

Q17

```
mysql> select emp_name, emp_salary from employee
    -> where emp_salary is NOT NULL;
+----------+------------+
| emp_name | emp_salary |
+----------+------------+
| Joey     |      47300 |
| Peter    |      60000 |
| Danial   |      55000 |
| Quagmire |     120000 |
| Stewie   |      62000 |
+----------+------------+
5 rows in set (0.00 sec)
```

Q18

```
mysql> select dept_name, sum(emp_salary) from employee
    -> group by dept_name
    -> having sum(emp_salary) > 100000;
+-----------+-----------------+
| dept_name | sum(emp_salary) |
+-----------+-----------------+
| SWAT      |          102300 |
| NYPD      |          242000 |
+-----------+-----------------+
2 rows in set (0.00 sec)
```

Q19

```
mysql> select distinct emp_name from employee;
+----------+
| emp_name |
+----------+
| Joey     |
| Peter    |
| Danial   |
| Quagmire |
| Stewie   |
+----------+
5 rows in set (0.00 sec)
```

Q20

```
mysql> SELECT emp_city, COUNT(DISTINCT dept_name) AS total_departments
    -> FROM employee
    -> GROUP BY emp_city;
+----------+-------------------+
| emp_city | total_departments |
+----------+-------------------+
| LA       |                 1 |
| NJ       |                 1 |
| NYC      |                 1 |
+----------+-------------------+
3 rows in set (0.05 sec)
```